

MUDIT GARG

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PROFILE

Software Engineer whose strengths lie in building **scalable, low-latency, production-grade systems** under real-world constraints, with experience in **distributed systems, full-stack development** and **AI-driven applications**, both as an individual contributor and as a team collaborator.

SKILLS

Languages:	Java 17, Python 3, SQL/NoSQL, JavaScript/TypeScript
Frameworks:	Spring (Boot, MVC, Hibernate), React (Redux, Next.js), Tailwind, Express, Flask, FastAPI
Databases & Tools:	MySQL, PostgreSQL, MongoDB, Docker, Git, Jira, Prisma ORM, n8n, GitHub Copilot
Cloud:	AWS (EC2, Lambda, S3, RDS, DynamoDB), Azure (AKS, DevOps), GCP (IAM, Firebase)
Concepts:	DSA, Sys Arch, CI/CD, Microservices, LLM, Gen AI, RESTful APIs, ORM, TDD, SOLID, DRY

EDUCATION

Masters in Engineering (M.A.I.). Major in Computer Engineering. Upper Second Class Honours (2:1) Grade.

Trinity College Dublin, Ireland

Sept 2023 – Oct 2024

Coursework: Information Retrieval and Search Optimization, Adaptive Applications, Software Engineering, Security and Privacy

Bachelors in Engineering (B.A.I.). Major in Computer Engineering. Upper Second Class Honours (2:1) Grade.

Participated in a 2+2 international credit transfer program based on merit awarded to the top 1% students of the batch. Final bachelor's degree was awarded by Trinity College Dublin

Trinity College Dublin, Ireland

Sept 2021 – Aug 2023

Coursework: Algorithm, Digital System Design, Microprocessor Systems, Signal and Systems, Probability and Statistics, Software Development Analysis, Computer Networks, Information Management for Engineers, Internet Applications, Database Management Systems, Deep Learning

Thapar Institute of Engineering and Technology, India

Aug 2019 – Aug 2021

Coursework: Data Structures, Operating Systems, Discrete Mathematical Structures, Machine Learning, Artificial Intelligence

Secondary High School Diploma. Major in Science. Minor in Computer Sciences

Sheiling House School, Kanpur, India

Till July 2019

EXPERIENCE

Hive: Real Time topic-centric Conversation Platform

Jan 2026 – Present

Independent Project | Production Deployment at <https://onhive.app>

- Engineered a real-time messaging ecosystem using **WebSocket-based fan-out architecture** and **buffered persistence** patterns, achieving **sub 200ms end-to-end latency for all core user interactions**.
- Owning end-to-end system design, deployment and scaling, validating system reliability and performance in a live production environment under constrained real-world usage and infrastructure.
- Optimized system performance using a cost-efficient architecture utilizing **Vercel/React for frontend** hosting and analytics, **Express.js/Render for backend** services, **Supabase/PostgreSQL for database** persistence.
- Currently handling **~35+ active users** with **~2,500+ interactions processed** in Alpha phase (graduating from Early Access Program in June), and with an estimated rollout to even larger communities by July.

Software Engineer (Freelance Consultant)

Jan 2025 – Present

Gravidad Systems S.L. (Las Palmas, Spain)

Remote

- Provided software development and technical consulting services across multiple client-driven initiatives like Gravidad Systems, utilizing **Power BI** and **Microsoft Excel** for document and impact analysis delivery.
- Reduced turnaround time for architectural analysis and frontend strategy deliverables by integrating AI-assisted code generation and validation workflows, utilizing **Claude Code for rapid prototyping and optimized workflows** to guide downstream engineering and technical decision-making.

Software Engineer (Intern)

Jan 2023 – Aug 2023

General Motors

Hybrid - Dublin, Ireland

- Collaborated on a **Java SE 17-based Spring Boot** monolith project with a team of 6, **optimizing performance by 96.4%** and **reducing technical debt by 23%**, streamlining the **CI/CD** pipeline by breaking it down to a **GraphQL-based** microservice architecture.
- Spearheaded the development of a new front-end architecture based on **Angular**, utilizing highly efficient, industry-standard UI/UX practices, **increasing user engagement by 15%** as compared to the previous iteration.
- Streamlined crucial **Power BI** reports in collaboration with senior engineers across time zones, **increasing productivity by 33%** through optimized screen real-estate and report consolidation.

PROJECTS

RaDAR: Resume and Job Description Analysis Resource
React 19, Vite, Redux, JSX, Flask, Gemini 2.5 Flash API, Google Cloud Platform (GCP)
Production Deployment at https://muditgarg48.github.io/radar_frontend/

- Developed an AI-powered analyzer that extracts key actionable insights from resumes and job descriptions, enhancing ATS compatibility and **reducing manual review and application tailoring time by 95% per application.**
- Incorporated an application tracker with smart status updates, **reducing job-search organization overhead by 80% and enabling users to track 200+ applications efficiently in one dashboard** without needing to juggle between platforms for related tasks.

Jun 2025

Portfolio Website with Retrieval Augmented Generation (RAG) based chatbot
React 18, JavaScript ES6, Python 3, Flask, ChromaDB, LangChain, Gemini 1.5 Flash API
Production Deployment at <https://muditgarg48.github.io>

- Designed and launched a dynamic portfolio website using **React** and **JavaScript**, resulting in a **30% increase in recruiter engagement** and an **18% rise in user adoption** due to improved user experience and streamlined access to professional details.
- Integrated a **Gemini API** and **ChromaDB**-powered AI chatbot, enabling query-driven portfolio exploration and **improving user interaction efficiency by 40%** through precise answers and guided navigation.

Oct 2024

Gesture recognition-based device control
Python 3, Jupyter, Bootstrap, Google Mediapipe
Production Deployment at https://github.com/muditgarg48/gesture_based_device_control

- Engineered a gesture recognition model using a custom **LSTM neural network**, **achieving 93% accuracy even on limited datasets.**
- Automated the development pipeline from project setup to IoT device communication using various **Python** packages like **Mediapipe**, **OpenCV**, **Tensorflow**, **NumPy** **Scikit-Learn** and **Home Assistant**.

May 2024

Candle in Dark
Flutter 3, Dart 3, Firebase, Google Cloud, XML
Production Deployment at https://muditgarg48.github.io/candle_in_dark_web/

- Ideated and developed a **cross platform responsive app** using **Flutter**, **Firebase** and **Google Cloud**, serving as a platform for all the info for the newly arriving students of Trinity College to know everything about Trinity and Ireland.
- Implemented several interwoven intricate systems including **user authentication using Google Sign-In**, **user authorization**, **role-based access**, **feedback collection system**, **light/dark mode**, **admin panel**, etc.

Mar 2024

CERTIFICATIONS

Oracle Cloud Infrastructure 2024 Generative AI Certified Professional	Oracle MyLearn	2024
Career Essentials in Generative AI	Microsoft, LinkedIn Learning	2024
Google AI Essentials	Coursera	2024
Programming Foundations of Visual Studio Code with Git	LinkedIn Learning	2022
Learning C	LinkedIn Learning	2022
Technical Support Fundamentals by Google	Google, Coursera	2021

ACHIEVEMENTS

- Apr 2024: Qualified for the finals by **securing a spot in the top 10** of **Gradireland Engineering Undergraduate of the Year** awards.
- Sept 2023: Awarded a **15% scholarship** to pursue my postgraduate degree in **Trinity College Dublin, Ireland**
- Sept 2019: Secured **1st Rank** in *Game of Codes*, a **live in-person coding competition** hosted by Thapar University’s OWASP society among a **pool of more than 200 participants**.
- July 2019: Scored a **92.6% percentile** in the highly acclimated Indian national entrance exam, **Joint Entrance Exam (JEE) Mains** attempted by **more than 12 lakh aspiring engineers**.